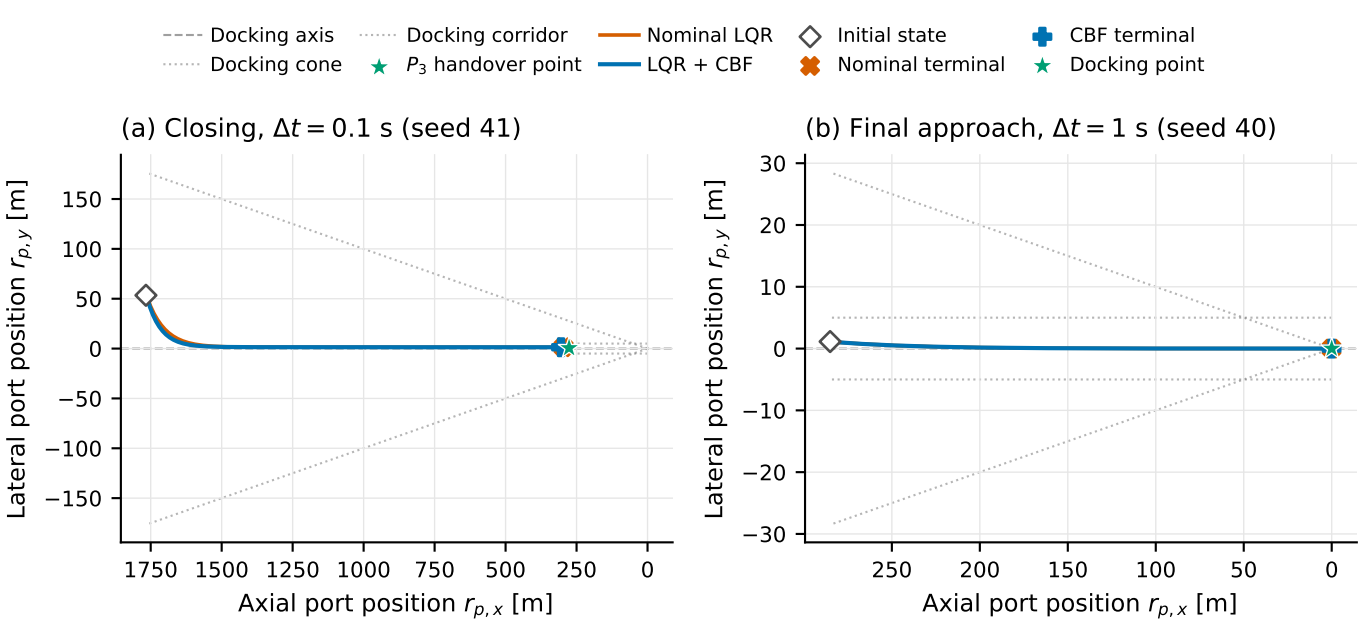
Docking axis
Docking cone
Docking corridor
P_3 handover point
Nominal LQR
LQR + CBF
Initial state
Nominal terminal
CBF terminal
Docking point
(a) Closing, $\Delta t = 0.1$ s (seed 41)
Lateral port position $r_{p,y}$ [m]
150
100
50
0
−50
−100
−150
1750
1500
1250
1000
750
500
250
0
Axial port position $r_{p,x}$ [m]
(b) Final approach, $\Delta t = 1$ s (seed 40)
Lateral port position $r_{p,y}$ [m]
30
20
10
0
−10
−20
−30
250
200
150
100
50
0
Axial port position $r_{p,x}$ [m]